Peiyao Wang

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Personal Statement

I am a graduate student majored in Electrical Engineering from Xidian University, China, and my advisor is Professor Weisheng Dong. My research interests include **image processing, deep learning and sparse representation**. Up to now, I have two papers accepted by an international journal (IEEE TPAMI) and an international conference (MMSP 2018), respectively.

Education

Xidian University, Xi'an, Shaanxi, China 09/2016-06/2019 (Expected)
M.S. student at the School of Artificial Intelligence
Majored in Electrical Engineering
Field: Image processing, Deep Learning, Sparse representation

Xidian University, Xi'an, Shaanxi, China 08/2012-06/2016
B.S. student at School of Electronic Engineering
Majored in Detecting Guidance and Control Technique
GPA, 3.7/4.0, Ranking in major: 2/38

Publications & Activities

Publications

- **Denoising Prior Driven Deep Neural Network for Image Restoration (first student author)**
  Accepted by IEEE Transactions on Pattern Analysis and Machine Intelligence
  Motivation: Design deep convolutional network supervised by traditional models, which are more explicable and robust.
  - Propose a denoising-based image restoration algorithm and design network modules supervised by the ite rative solution.
  - Jointly optimize the denoisers and the BP modules through end-to-end training.
  - Results achieve state-of-the-art on image restoration tasks such as denoising, super-resolution and deblurring.

- **ConvCSNet: A Convolutional Compressive Sensing Framework Based on Deep Learning**
  Accepted by International Workshop on Multimedia Signal Processing
  About: A novel two-branch convolutional neural network to sense the whole image using a set of convolutional filters and reconstructs images.
  - implement the sensing by convolving the input image with a set of random filters followed by

subsampling

- reconstruct the input image from the linear measurements with a novel CNN containing two branches, which is based on the sparsity-based reconstruction model

Activities

- China Multiply Media(ChinaMM2018) 09/2018
- 2018 International Conference on Pattern Recognition(2018 ICPR), *oral report* 08/2018
- ShanghaiTech Symposium on Information Science and Technology(SIST2018) 07/2018
- 2018 Vision And Learning Seminar (2018VALSE) 04/2018

Awards & Honors

Awards

- BeiDou-Cup China Adolescents Science Technology Invention Contest, *Excellence* 05/2015
- **Interdisciplinary Contest In Modeling of America**, *Meritorious Winner* 04/2015
- China Undergraduate Mathematical Contest in Modeling, *Second Prize* 12/2014
- Mathematical Contest in Modeling of Xidian University, *Second Prize* 06/2014

Honors

- First Class Graducate Student Scholarships(Top10%) 2016-2018
- Outstanding Graduate Student(Top10%) 10/2017
- First Class Ungranduate Student Scholarships(Top10%) 2013-2015

Skills

Programming Language: Python, Matlab, C

Tools: Tensorflow, Caffe, MatConvNet